ANJANA ANAND

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EDUCATION

University of Colorado Boulder | MS Data Science | GPA: 3.83/4.0 Expected: May 2027
Coursework: Statistics, Data Mining, Data economics, Exploratory Data Analysis, Bootstrapping, Data Structures, Cyber security

Shiv Nadar University | BTech Artificial Intelligence and Data Science Graduated: May 2025
Coursework: Natural Language Processing, Computer Vision, Machine Learning, Time-Series Analysis, Data Structures and Algorithms, Quantitative Modeling, Database Management Systems.

TECHNICAL SKILLS

General-purpose: Python, R, C, C++, JavaScript, MATLAB, Bash
Data Science: SQL, Pandas, Numpy, Excel, Statistics
ML: Scikit-Learn, TensorFlow, LSTM, PyTorch, YOLOv8, ResNeXt, MAMBA, Pegasus, LLM
Data Visualization: Tableau, PowerPoint, Streamlit
Computer Vision: OpenCV, cvzone
Data Quality: Great Expectations
Frontend Frameworks: React , HTML
Version Control: GitHub, Bitbucket

EXPERIENCE

Positive Integers - Data Science Intern May 2026 - Present
Project: Return Risk Scoring & Win/Loss Analysis | Python, LightGBM, XGBoost, CatBoost, HistGradientBoosting
- Designed a weighted behavioral risk-scoring framework (Return Probability Score) to attribute distributor-level returns to individual retail outlets for an FMCG client, engineering **7 features** combined via quantile-rank normalization.
- Built a cross-correlation lag-analysis pipeline per SKU subcategory quantifying sales-to-return lag, **feeding a lag feature into the risk score** for subcategories with significant lag signal.
- For a B2B telecom client, ran statistical EDA across **9 opportunity attributes** to identify win/loss drivers despite a **99% win-rate imbalance**, then built and backtested two-stage CatBoost/XGBoost/LightGBM models plus a HistGradientBoosting pipeline for sales-cycle duration.

University of Colorado Boulder, Leeds School of Business - Graduate Teaching Assistant Spring 2026
Course: MS Business Analytics Experiential Projects
- Attended weekly industry partner meetings to align academic project deliverables with real-world business needs.
- Graded project reports and provided structured feedback for graduate business analytics students.
- Conducted review sessions to reinforce course concepts and support student project execution.

Crayon Data - Data Quality Intern January 2025 - June 2025
Project: Data Integrity Pipeline | Python, Great Expectations, Pandas, SQL
- Developed and deployed an automated data quality pipeline across 5+ internal datasets, **improving data consistency by 35%.**
- Integrated validation checks to detect and auto-correct 1,000+ data anomalies, **reducing manual data cleaning time by 80%.**
- Accelerated downstream analytics workflows by automating data validation and quality management, increasing overall processing **efficiency by 50%.**

Indian Institute of Technology Madras - Research Assistant May 2024 - August 2024
Project: Tennis ball trajectory analysis | Python, YOLOv8, OpenCV, Computer Vision
- Implemented YOLOv8 based object detection to identify ball movement across 20+ rally sequences in real-time with **95% precision.**
- Developed a high-accuracy tracking system for 20 datasets **improving inference speed** for computer vision applications by 30%.
- Developed robust tracking algorithms for tennis ball trajectories furthering the benchmark results in the field.

Chistats Labs - Data Science Intern May 2023 - July 2023
Project: Stock price forecasting using time-series modeling | Python, MATLAB, Time-Series Analysis, Quantitative Modeling
- Performed feature engineering and time-series modeling on 5 years of market data accelerating **forecast accuracy by 18%.**
- Applied statistical analysis techniques including ARIMA modeling and regression to enhance data-driven decision-making capabilities.
- Identified volatility clusters and trend patterns to enhance predictive stability and ensure consistent model performance across varying market conditions.

PROJECTS

Occupational Wage Analysis | Python, Streamlit, Pandas 2026
- Analyzed **20 years (2005–2024)** of BLS wage data spanning **800+ occupations** across all 50 states (**739K records**), engineering 79 lag-based features from wage and employment history.
- Built a Random Forest regression model achieving $\mathbf{R^2}$ **of 0.97** (RMSE ~$5,234) and a Decision Tree classifier reaching **90% accuracy** on wage-tier classification; deployed results via an interactive Streamlit dashboard.

Distracted Driving Detection | Python, ResNeXt50, MAMBA, AI-City Dataset 2025
- Designed a vision-based model to detect distracted driving behaviors from in-vehicle footage improving **accuracy benchmark by 72%.**
- Trained and fine-tuned benchmark models like ResNeXt50 and MAMBA architectures enhancing generalization across viewpoints.

Play Summarizer | Python, Pegasus, Transformers 2024
- Programmed an NLP pipeline using Pegasus to convert dialogues into structured narratives for **200+ play scripts.**
- Generated coherent summaries in transformer-based NLP model benchmarks **optimizing evaluation scores by 30%.**

Basketball Shot prediction | Python, OpenCV, cvzone, NumPy 2024
- Designed a basketball analytics system using OpenCV, cvzone, and NumPy to track ball trajectories across **20+ videos.**
- Used a regression-centric trajectory analysis to achieve shot outcome **prediction accuracy of 90%.**

Cryptocurrency Volatility Prediction | Python, LSTM, NLP 2023
- Engineered an LSTM model to forecast cryptocurrency volatility using 50,000+ social media posts, **improving prediction accuracy by 21%.**
- Mitigated class imbalance with advanced sampling techniques, **achieving 81.3% overall model accuracy.**